

BMIA - Practical case 2: Methodology for Diabetic Macular Edema segmentation

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I. INTRODUCTION

The goal of this practice is to create a fully automated system for the segmentation of pathological fluid regions in OCT images, in order to the accurate diagnosis and monitoring of this relevant ocular pathology in patients and support the clinicians. The image set provided consists of 50 OCT scans (left Fig. 1) with pathological fluid regions and 50 Ground Truth images (right Fig. 1) with annotations by clinical experts. Images are in *jpg* format and have different dimensions, all of them with rectangular shapes.

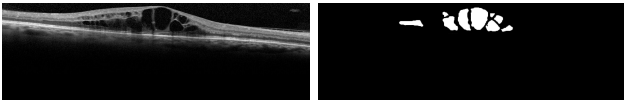


Fig. 1. Left: OCT scan with pathological fluid regions. Right: mask image with annotations by experts.

II. METHODOLOGY

The process and chosen techniques for the current methodology were divided into 4 parts: image pre-processing, DME segmentation methods and results of each. Each section is explored below:

A. Pre-processing of the image

In order to homogenize all images, a small pre-processing was done. The transformations followed were:

- Resizing all images to 512 x 512 with a liner interpolation from *cv2* package.
- For the mask only one channel is taken and transformed into grayscale.

B. Dataset and evaluation metrics

The data was split, initially, into train (70%) and test (30%) sets. In order to measure the stability of the model a cross-validation k-fold with 10 splits was done for each method implemented considering the train set; the test was used only for the final predictions.

Regarding the metrics for the performance of the models, *Dice* and *Iou* scores were gotten.

- **Dice score**, also known as the Dice Similarity Coefficient, is a measure that can be used to evaluate the similarity

between a predicted segmentation mask and the ground truth segmentation mask. The Dice score ranges from 0, indicating no overlap, to 1, indicating perfect overlap.

- **Intersection over Union (IoU)** is used to evaluate object detection performance by comparing the ground truth bounding box to the predicted bounding box. The metric value interpretation is straightforward. The bigger the overlapping, the higher the score, the better the result.

C. DME Segmentation algorithms

Below, each of the models tried for the segmentation task is explained in more detail.

1) **Attention UNet**: U-net architecture (Figure 2) is very well known for biomedical image segmentation. Its architecture can be broadly thought of as an encoder network followed by a decoder network. Unlike classification where the end result of the deep network is the only important thing, semantic segmentation not only requires discrimination at the pixel level but also a mechanism to project the discriminative features learnt at different stages of the encoder onto the pixel space.

Attention, in the context of image segmentation, is a way to highlight only the relevant activations during training. This reduces the computational resources wasted on irrelevant activations, providing the network with better generalisation power. Essentially, the network can pay “attention” to certain parts of the image.

The parameters and hyperparameters used were:

- *optimizer* = *RMSprop*
- *learning rate* = $1e-3$
- *loss* = *focal tversky loss*
- *metrics* = [*iou*, *iou thresholded*]
- *output layer activation* = “*swish*”
- *epochs* = 300
- *callbacks* = [*ModelCheckpoint*, *ReduceLROnPlateau*, *EarlyStopping*]

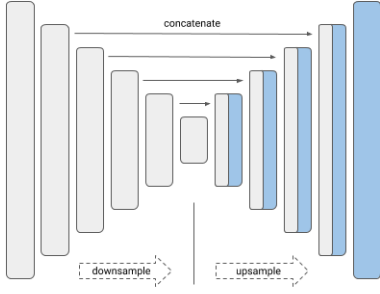


Fig. 2. U-Net architecture

2) **UNet with backbone:** The "backbone" refers to the feature extracting network which is used within the UNet architecture. This feature extractor is used to encode the network's input into a certain feature representation. And later, decoder the image into the segmentation mask expected. The library used for this and the next architecture is available https://github.com/qubvel/segmentation_models.

The parameters and hyperparameters used were:

- *optimizer* = *RMSprop*
- *learning rate* = $1e-3$
- *loss* = *total loss*
- *metrics* = [*iou score*, *fscore*]
- *output layer activation* = *sigmoid*
- *backbone* = *resnet 34*
- *encoder weights* = *imagenet*
- *epochs* = 20
- *callbacks* = [*ModelCheckpoint*, *ReduceLROnPlateau*, *EarlyStopping*]

3) **LinkNet with backbone:** U-Net uses skip connections from the encoder to the decoder before each size-reduction step. There is a skip connection that takes the features from the encoder and concatenates them to the decoder thus giving the decoder enough context to generate a proper segmentation mask. If we replace the operation of concatenation with the addition we get the network called LinkNet (Figure 3).

The parameters and hyperparameters used were:

- *optimizer* = *RMSprop*
- *learning rate* = $1e-3$
- *loss* = *total loss*
- *metrics* = [*iou score*, *fscore*]
- *output layer activation* = *sigmoid*
- *backbone* = *inception v3*
- *encoder weights* = *imagenet*
- *epochs* = 20
- *callbacks* = [*ModelCheckpoint*, *ReduceLROnPlateau*, *EarlyStopping*]

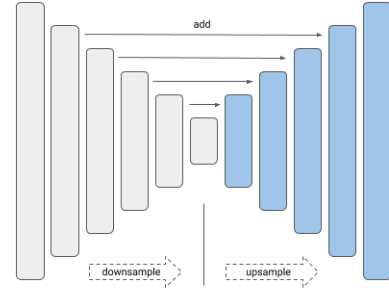


Fig. 3. Link-Net architecture

D. Results

In Table I, the results for Attention Unet and LinkNet with InceptionV3 backbone are shown. In the second column, there are the values of the mean and standard deviation of the IoU score for the 10 splits for the cross-validation; in the third column, we do have the performance of each method in the test dataset via Dice and IoU scores. Clearly, the LinkNet with Inception V3 backbones and encoder weights from imagenet is showing better results than the Attention U-Net, with almost 3 points above for both metrics in the predictions. In Figure 5 and Figure 6, we might visualize some examples of the prediction masks for Attention Unet and LinkNet respectively.

In Figure 4, there are two examples of different data splits with the performance's history for each epoch of training. Looking into that and some of the segmentation predictions, one conclusion gotten was that, probably, there are some patterns more complex than others and the algorithms are having problems to detect.

TABLE I

Method	Validation		Test	
	mean	std	Dice	IoU
Attention Unet	0.554	0.119	0.576	0.454
LinkNet inception v3 backbone	0.640	0.110	0.856	0.769

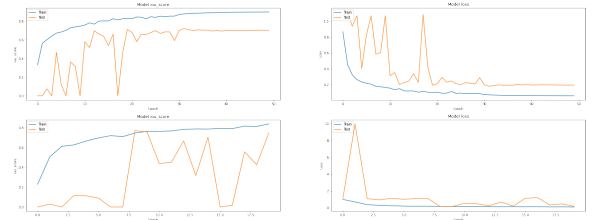


Fig. 4. History of IoU score and loss for different splits

III. DISCARDED EXPERIMENTS

Finally, one of the models describes before, UNet with Resnet34 backbones, had very bad results, nearly to zero in

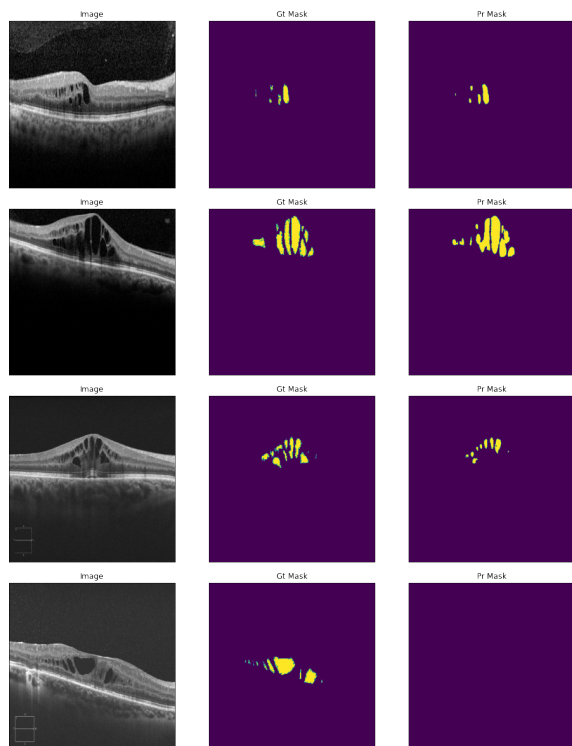


Fig. 5. Visualization of Attention UNet results in test

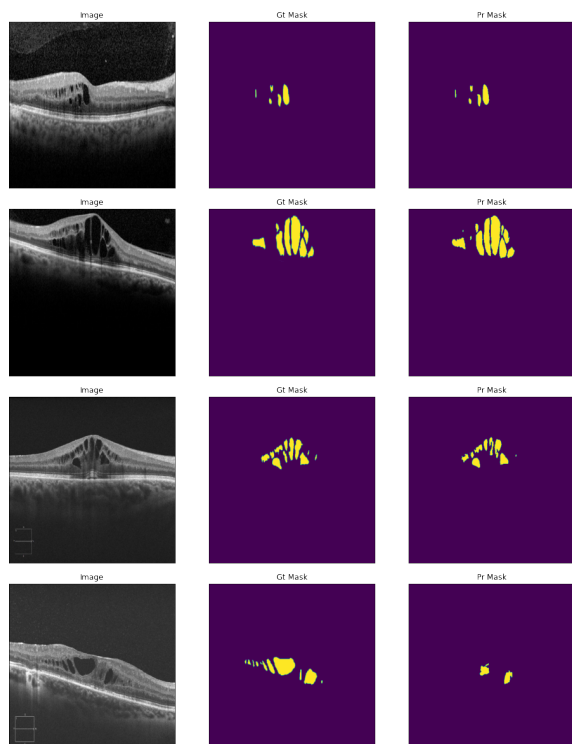


Fig. 6. Visualization of LinkNet with InceptionV3 backbone results in test

is shown as a discarded experiment. Also, it is important to mention that a lot of different combinations of the parameters and hyperparameters were used for experiments until obtain good results.

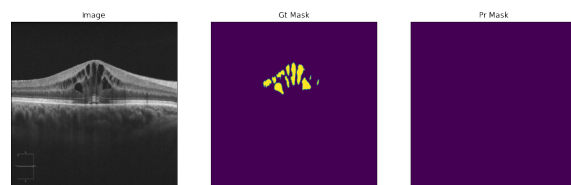


Fig. 7. Visualization of UNet with Resnet34 backbone results in test

the metrics, an example of it is shown in Figure 7; that is the reason why it wasn't included in the table of results but it